

Chapter 6 - Class B Subnetting

1. What is a Class B IP Address?

A **Class B IP address** is used for **medium to large-sized networks**  .

Class B Range

- **128.0.0.0 to 191.255.255.255**

Default Subnet Mask

- **/16**
- **First 16 bits** → Network part
- **Last 16 bits** → Host part

Daily Life Example:

- Think of a university campus  where each department has many computers.
- Class B allows many sub-networks with many hosts.

2. Understanding the Given IP

◆ IP Address:

172.168.10.0/22

- 172.x.x.x → **Class B**
 - Default Class B mask = **/16**
 - Given mask = **/22**
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3. What Does /22 Mean?

- Total bits in IPv4 = 32
- Network bits = 22
- Host bits = $32 - 22 = 10$

 Extra bits added to network = **Borrowed bits**

4. Borrowed Bits Calculation

- Default mask = /16
- Given mask = /22

Borrowed Bits

$$22 - 16 = 6 \text{ bits}$$

 These 6 bits are borrowed from the **host portion** and added to the **network portion** 

5. How Many Subnets?

Formula

$$\text{Number of Subnets} = 2^n$$

- $n = \text{borrowed bits} = 6$

Result

$$2^6 = 64 \text{ subnets}$$

Why needed?

- To divide one big organization network into multiple departments 
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6. IP Addresses per Subnet

◆ Remaining Host Bits

$$32 - 22 = 10 \text{ bits}$$

Formula

$$\text{Total IPs per Subnet} = 2^{10} = 1024$$

✓ Usable Hosts

$$1024 - 2 = 1022$$

? Why Subtract 2?

- First IP → **Network Address** 
- Last IP → **Broadcast Address** 

7. Finding Network ID Range (Very Important 🔥)

Key Rule

- Check **which octet is affected by subnetting**
- In **/22**, subnetting happens in the **3rd octet**

Block Size Calculation

- Borrowed bits in 3rd octet = **2 bits**
 - Block size = **$2^2 = 4$**
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8. Network Ranges for 172.168.10.0/22

Each subnet increases by 4 in the 3rd octet 🙋

◆ First Few Subnets

Subnet No.	Network ID	Broadcast Address
1	172.168.0.0	172.168.3.255
2	172.168.4.0	172.168.7.255
3	172.168.8.0	172.168.11.255
4	172.168.12.0	172.168.15.255
...	...	...
64	172.168.252.0	172.168.255.255

🙋 Total **64 networks** will be created ✓



Daily Life Analogy

- Each **subnet** = one department 🏢
- Network ID = department name board
- Broadcast = office announcement system 🔔



9. Example 2: Class B Subnetting – 172.168.0.0/21

◆ Step 1: Identify Class

- 172.x.x.x → Class B
- Default mask = /16

◆ Step 2: Borrowed Bits

$$21 - 16 = 5 \text{ bits}$$

◆ Step 3: Number of Subnets

$$2^5 = 32 \text{ subnets}$$

◆ Step 4: Host Bits & Hosts

- Host bits = $32 - 21 = 11 \text{ bits}$

$$\text{Total IPs} = 2^{11} = 2048$$

$$\text{Usable Hosts} = 2046$$

◆ Step 5: Block Size & Network Range

- Subnetting octet = **3rd octet**
- Borrowed bits in 3rd octet = **3 bits**

$$\text{Block size} = 2^3 = 8$$

◆ Network Ranges

	Subnet Network ID	Broadcast
1	172.168.0.0	172.168.7.255
2	172.168.8.0	172.168.15.255

Subnet	Network ID	Broadcast
3	172.168.16.0	172.168.23.255
...	...	...
32	172.168.248.0	172.168.255.255

Key Points to Remember

- Class B default mask = /16
 - Borrowed bits decide number of subnets
 - Block size helps find network ID
 - Always subtract 2 for usable hosts
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◆ 1 Chapter Summary

Class B subnetting is used to divide **medium to large networks** into multiple smaller and manageable sub-networks 🏢🏠.

A **Class B IP address**:

- Ranges from **128.0.0.0 – 191.255.255.255**
- Has a **default subnet mask /16**
- First **16 bits = Network**
- Last **16 bits = Host**

By using **CIDR notation** like /22 or /21, we:

- Borrow bits from the host portion
- Increase the number of networks (subnets)
- Control the number of hosts per subnet

Key concepts include:

- **Borrowed bits**
- **Number of subnets = 2^n**
- **Hosts per subnet = $2^n - 2$**
- **Block size** to find network ranges

👉 This chapter explains **how large organizations logically divide their networks** for efficiency, security, and scalability 🌐🔒

◆ 2 Chapter Conclusion 🎯

Class B subnetting is essential for **real-world enterprise networks** like:

- Universities 🎓
- Corporate offices 🏢
- Data centers 💻

By mastering this chapter, you learn:

- How to divide a big IP block into departments
- How CIDR gives flexibility beyond classful limits
- How to calculate subnets, hosts, and ranges confidently

💡 An interviewer can clearly see that:

You understand **both theory and practical subnet planning**

👉 Strong Class B subnetting = Strong networking fundamentals 💪 🌍

◆ 3 Detailed Mind Map

Class B Subnetting

- |
- |— Class B Overview
- | |— Range: 128.0.0.0 – 191.255.255.255
- | |— Default Mask: /16
- | |— 16 Network Bits
- | |— 16 Host Bits
- |
- |— CIDR Concept
- | |— /21, /22, etc.
- | |— Flexible subnetting
- | |— Borrowing host bits
- |
- |— Borrowed Bits
- | |— Given mask – Default mask
- | |— Decide number of subnets
- |
- |— Calculations
- | |— Subnets = 2^n
- | |— Hosts = $2^n - 2$
- | |— Block size
- |
- |— Block Size Logic

- | └— Identify subnetting octet

- | └— Use borrowed bits

- | └— Increment network IDs

- |

- | └— Example /22

- | └— 64 subnets

- | └— 1022 hosts each

- | └— Jump = 4 (3rd octet)

- |

- | └— Example /21

- | └— 32 subnets

- | └— 2046 hosts each

- | └— Jump = 8 (3rd octet)

- |

- | └— Key Rules

- | └— Always subtract 2

- | └— Network & Broadcast reserved

- | └— CIDR removes class limitations

◆ Q & A

? Q1. What is a Class B IP address?

Answer:

A Class B IP address is used for **medium to large networks** and ranges from **128.0.0.0 to 191.255.255.255**.

It has a default subnet mask of **/16**.

? Q2. How many network and host bits are there in Class B?

Answer:

- Network bits = 16
- Host bits = 16

This allows many subnets and many hosts per subnet 🧠.

? Q3. What is the default subnet mask of Class B?

Answer:

The default subnet mask of Class B is **255.255.0.0** or **/16**.

? Q4. What does CIDR /22 mean?

Answer:

/22 means:

- 22 bits are used for the network
- 10 bits are used for hosts

It allows flexible subnetting beyond default class limits 🎯.

? Q5. How do you calculate borrowed bits?

Answer:

Borrowed bits =

👉 **Given mask – Default mask**

Example:

/22 – /16 = **6 borrowed bits**

? Q6. How many subnets are created in /22?

Answer:

Formula:

👉 **Subnets = 2^n**

Where n = borrowed bits = 6

So, **$2^6 = 64$ subnets**

? Q7. How many hosts per subnet in /22?

Answer:

Host bits = $32 - 22 = 10$

Total IPs = $2^{10} = 1024$

Usable hosts = **1022**

? Q8. Why do we subtract 2 from total IPs?

Answer:

Because:

- 1 IP is reserved for **Network ID**
- 1 IP is reserved for **Broadcast ID**

These cannot be assigned to devices **✗**.

? Q9. What is block size in subnetting?

Answer:

Block size is the **jump value** between network IDs.

It helps find subnet ranges quickly ⚡ .

? Q10. In /22, which octet is used for subnetting?

Answer:

In /22, subnetting occurs in the **3rd octet**.

? Q11. What is the block size for /22?

Answer:

Borrowed bits in 3rd octet = 2

Block size = $2^2 = 4$

? Q12. Explain subnet ranges for 172.168.10.0/22

Answer:

Networks increase by **4 in the 3rd octet**:

- 172.168.0.0 – 172.168.3.255
- 172.168.4.0 – 172.168.7.255
- 172.168.8.0 – 172.168.11.255

Total **64 subnets** ✓

? Q13. Explain subnetting of 172.168.0.0/21

Answer:

- Borrowed bits = 5
- Subnets = 32
- Host bits = 11

- Usable hosts = 2046
 - Block size = 8 (3rd octet)
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? Q14. Why is Class B suitable for large organizations?

Answer:

Because it:

- Supports many subnets
 - Supports thousands of hosts
 - Is scalable and efficient for enterprises 🏢🌐
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? Q15. What happens if subnetting is not used?

Answer:

Without subnetting:

- Network traffic increases 🚦
 - Security decreases 🗝️
 - Network becomes slow and unmanageable 🚨
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? Q16. How do you quickly identify Class B?

Answer:

If the **first octet is between 128 and 191**, it is **Class B**.

? Q17. What is the real-life analogy of Class B subnetting?

Answer:

A university campus:

- Campus = Network
- Departments = Subnets
- Students/PCs = Hosts 🎓💻